

Technology Stack

Date	22 June 2026
Team ID	
Project Name	EduGenie
Maximum Marks	2 Marks

Technology Stack:

The EduGenie platform uses a carefully selected technology stack to support the development of an AI-powered educational assistant. The chosen technologies ensure that the application is reliable, maintainable, scalable, and capable of delivering personalized learning experiences. The technology stack includes programming languages, frameworks, front-end and back-end tools, databases, and APIs that work together to provide seamless interaction between users and AI services.

The table below details the technologies used for each component of the system architecture along with a brief justification for their selection.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Vanilla JavaScript, Jinja2	Provides responsive structure, modern styling (glassmorphism), interactive DOM logic, and server-side page templates.

2	Backend / Server-Side	FastAPI, Uvicorn, Python 3.10+	Fast asynchronous processing, native type validations, and automatic Swagger docs.
3	Database / Data Storage	JSON file-based storage	Lightweight, schema-free representation of users, settings, roadmaps, and history logs.
4	Cloud / Hosting / Deployment	Local deployment	Hosted locally using Uvicorn server, suitable for offline setups and simple deployment.
5	Version Control & CI/CD	Git, GitHub	Used for collaborative development, task branching, and version tracking.

6	Third-Party APIs / Other Tools	Google Gemini 1.5 Pro API & PyPDF2 / docx	Powering AI educational Q&A, quiz generations, study path roadmaps, summaries, and PDF/DOCX document text extraction for RAG.